What's for dinner? A Comparison of Gulf of Maine Prey Energy Densities

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Introduction

Forage fishes and other small-bodied pelagic prey connect and disperse energy across food webs through movement and consumption by predators[1,2]

Higher prey energy density (ED; energy stored per unit mass) = Higher value of prey to predators[2]

Higher availability of energy dense prey = Higher ecosystem productivity and resilience to ecosystem stressors[1,3]

Recent climate-mediated changes in prey availability and energy content could alter Gulf of Maine (GOM) energy flows and impact ecosystem resilience[2,4]

Studying common prey items and changes in energy content is essential to understanding how these energy flows and ecosystem dynamics will shift

Objectives:

1) Identify which prey species have the highest energy content
2) Compare our prey energy density values (2021-2026) to Hermann et al.'s prey energy density database values (1973-2024)[3]

Methods

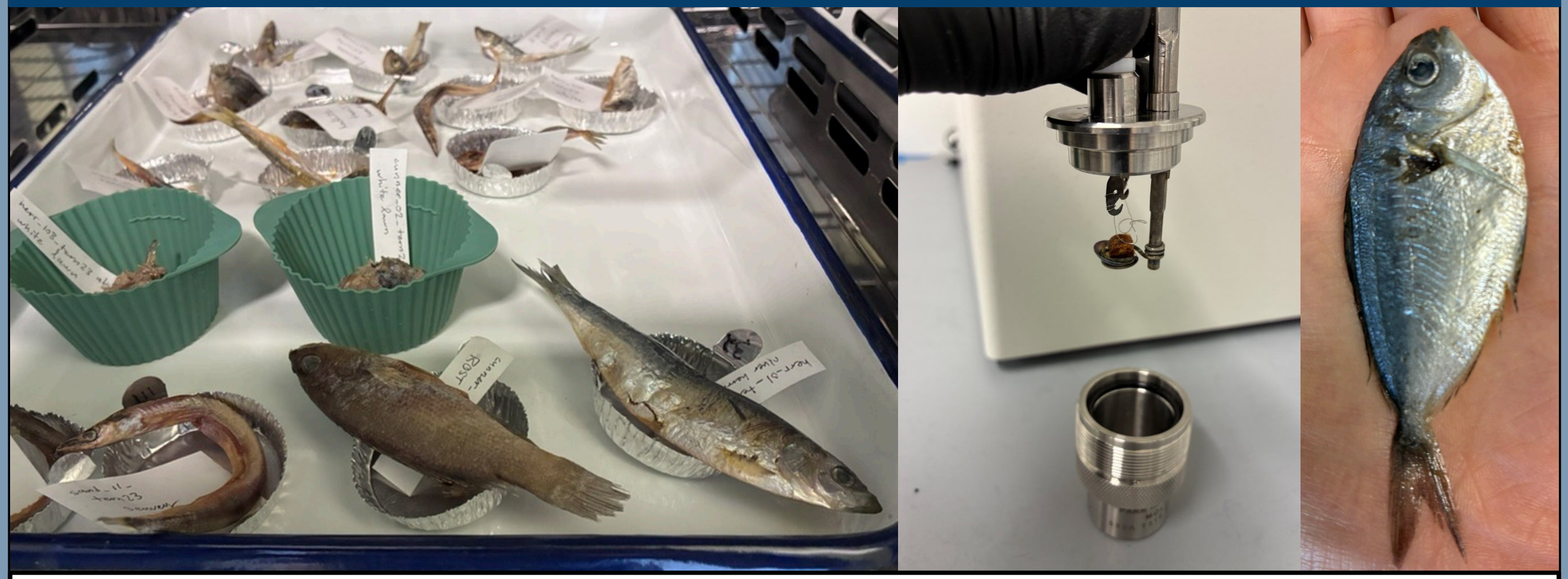

Figure 1a (left): Forage fishes in the drying oven; **Figure 1b (middle):** A pellet with a fuse attached entering the bomb for bomb calorimetry; **Figure 1c (right):** A butterfish sample collected after being dropped by a tern (White Island, NH).

We use bomb calorimetry to collect prey ED data:

1) Collect intact samples of prey found in cod, striped bass, and common tern diets (Figure 3)
2) Collect morphometric data and dry samples at 60°C until consecutive stable dry weights (Figure 1)
3) Homogenize sample using mortar and pestle and press into pellets
4) Weigh pellets and combust using the bomb calorimeter (Figure 2)
5) Kruskal Wallis & Dunn tests to identify differences in mean species ED
6) Comparison of mean ED values to ED database (Hermann et al.)[3]

Preliminary Results

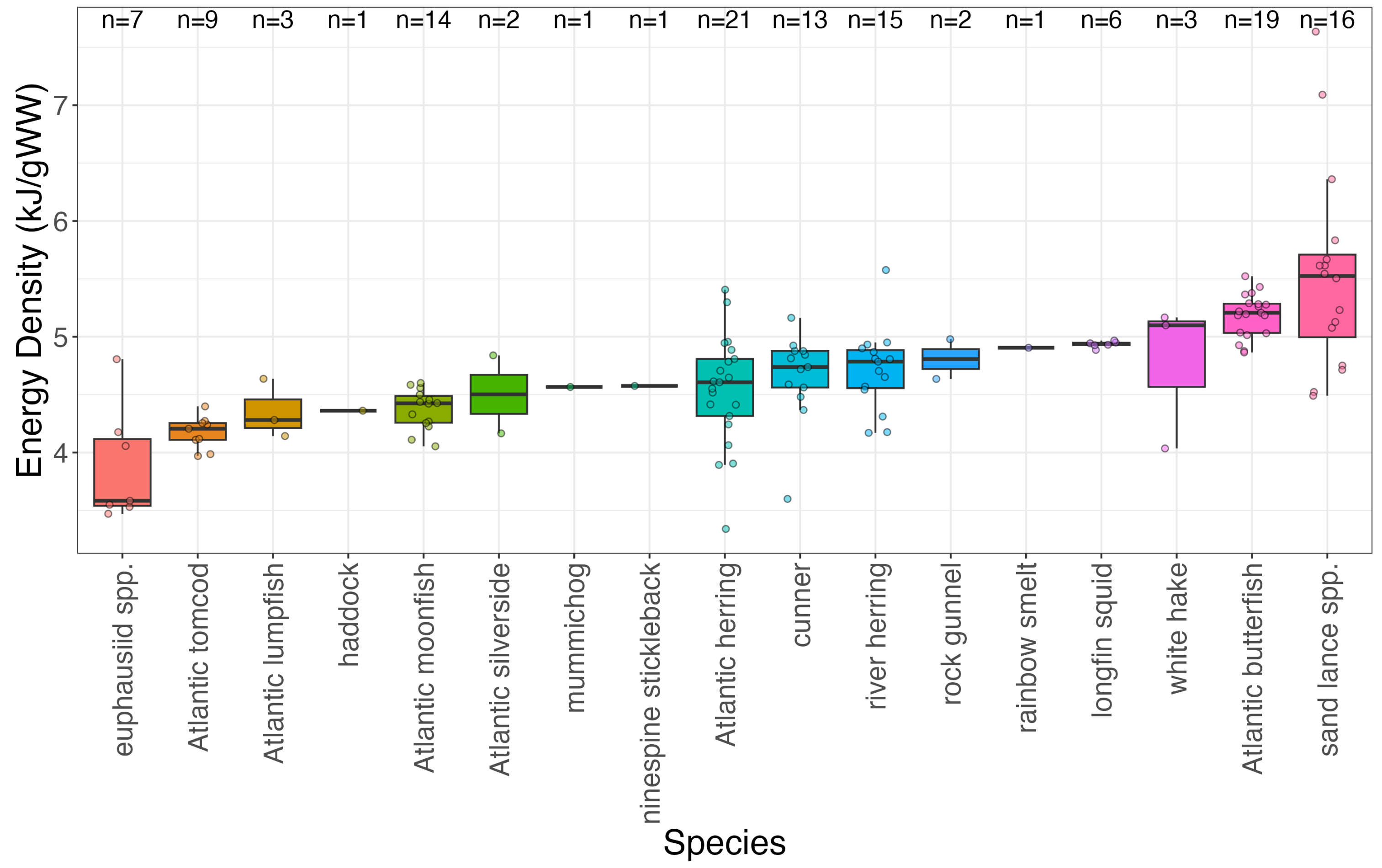

Figure 2: A comparison of 17 Gulf of Maine forage fish energy densities (kilojoules per gram wet weight). River herring encompasses blueback herring (*Alosa aestivalis*) and alewife (*Alosa pseudoharengus*). Sand lance encompasses American sand lance (*Ammodytes americanus*) and Northern sand lance (*Ammodytes dubius*).

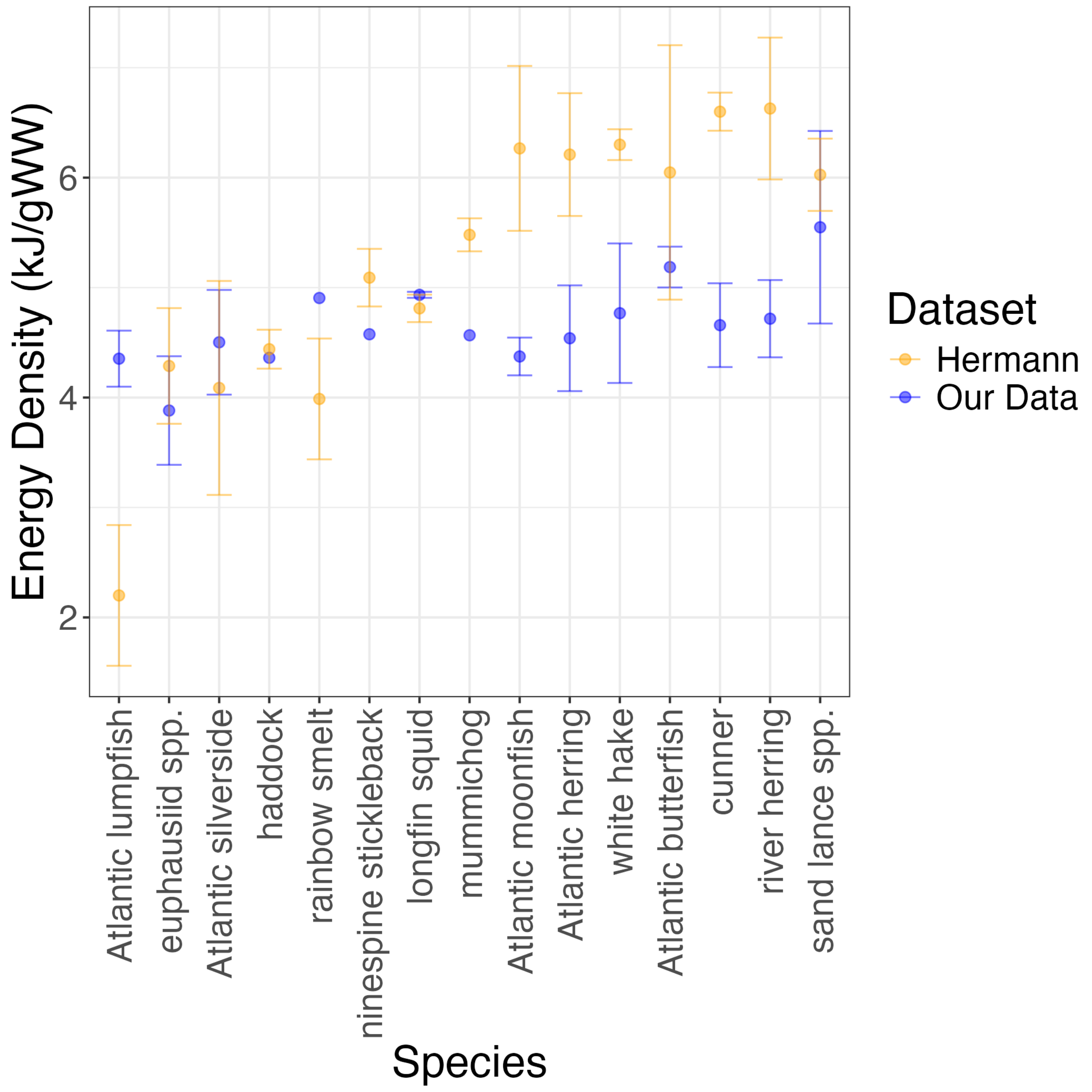

Figure 3: A comparison of 15 species energy density values from our dataset (2021-2026) to the largest energy density database (Hermann et al.[3]; 134 total sources spanning from 1973-2024). Hermann et al. data for lumpfish represents the Cyclopteridae family. River herring encompasses blueback herring (*Alosa aestivalis*) and alewife (*Alosa pseudoharengus*). Sand lance encompasses American sand lance (*Ammodytes americanus*) and Northern sand lance (*Ammodytes dubius*).

Preliminary Results Continued

- Hermann et al. reported higher ED values than our data for 11 of the 15 prey compared (Figure 5)
- Sand lance (*Ammodytes spp.*), white hake (*Urophycis tenuis*), and Atlantic butterfish (*Peprilus triacanthus*) are the most energy-rich prey that we sampled (2021-2026) while euphausiid spp., Atlantic tomcod (*Microgadus tomcod*), and Atlantic lumpfish (*Cyclopterus lumpus*) are the least energy-rich prey that we sampled (Figure 4)
- Atlantic butterfish and sand lance had significantly higher EDs than the euphausiid spp., Atlantic moonfish (*Selene setapinnis*), and Atlantic tomcod (Kruskal Wallis test with post-hoc Dunn test; $p < 0.001$)

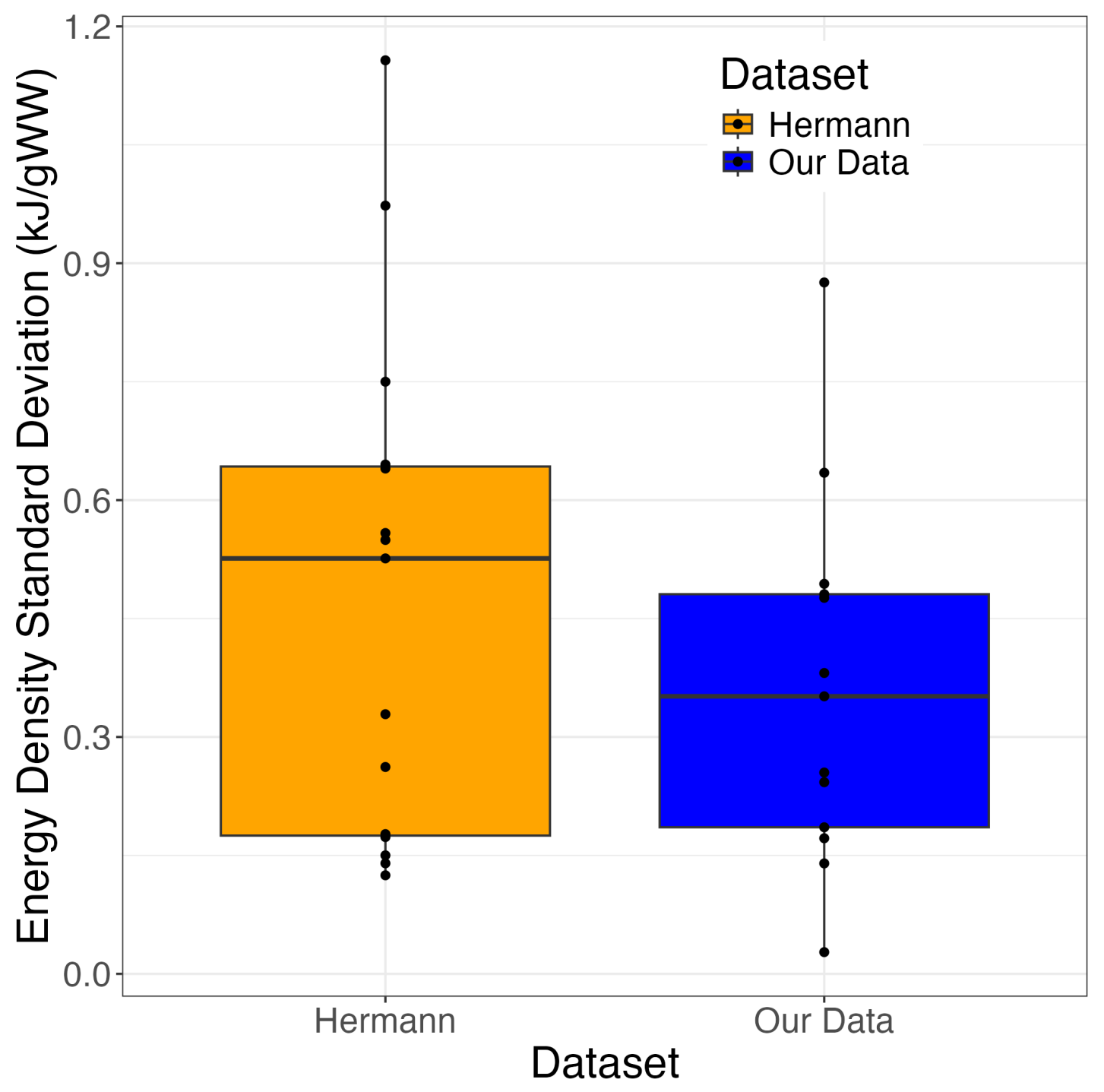

Figure 4: A comparison of standard deviations for mean energy densities between literature (Hermann et al.) and our results.

Takeaways

- Ongoing changes in prey species abundance in the GOM could alter the availability of energy dense prey to marine predators and impact predator foraging effort
- Lower ED values observed in our study, compared to literature[3], could indicate lower ecosystem productivity due to ocean warming trends
- High standard deviation in ED observed within species can likely be attributed to variations in life stages and environmental conditions across samples
- These data can be used to understand the bioenergetic effects of continued ocean warming on prey and their predators in the GOM

Acknowledgements

References